CHAPTER 14

Random variables, concentration and law of large numbers

S62, S64, S73, S74, S77

Before you start · Laws and dispersion

Without a calculator, in six minutes. Respond briefly. Mobilize random variables and Inequalities. Then choose the targeted review indicated under the questions.

1. Give the expectation of a law that is 0 or 1 with probabilities 0.7 and 0.3.

2. Calculate $V(X)$ for a Bernoulli parameter p.

3. Translate $|x-2| \geq 1$ without absolute value.

4. What is the opposite inequality?

5. Calculate the average of 1,2,3,4.

6. Is an average always an integer?

Responses

0,3 ; $p(1-p)$; x not more than 1 or at least 3 ; $1 < x < 3$; 5/2; no.

Targeted review after the diagnostic check

Repeat only the missed questions. For each question: **1.** Weight each two values by its probability. **2.** Calculate E(X2)-E(X)2 for a variable 0/1; R51. **3.** Draw Remote numbers at least 1 of 2. **4.** Complement the two half-rights, excluded limits. **5.** Sommer then divide by the number of values. **6.** Test the two values 0 and 1. R are used as support with the teacher, even if their collective ritual is planned more late.

Objectives

Calculate expectation and variance of a sum and average; apply Bienaymé-Chebychev ; determine a sample size.

Course notes

Amounts and changes

For a finite random variable X, taking the values x_i with probabilities p_i,

$$E(X) = \sum_i x_i p_i, \qquad V(X) = \sum_i (x_i - E(X))^2 p_i = E(X^2) - E(X)^2.$$

The standard deviation is $\sigma(X) = \sqrt{V(X)}$. Developing the square in $V(aX+b)$, the lag b seliminate and the factor a is raised squared. For finite random variables X, Y and Real a, b,

$$E(X+Y) = E(X) + E(Y), \quad E(aX+b) = aE(X) + b, \quad V(aX+b) = a^2 V(X).$$

The linearity of expectation does not require independence. X, Y are independent, $V(X+Y) = V(X) + V(Y)$ (admitted). Without this hypothesis, this formula cannot be applied.
$aX+b$ value $|a|\sigma(X)$.

Example of addiction. If $Y = X$, then $V(X+Y) = V(2X) = 4V(X)$, not 2 $V(X)$ as soon as $V(X) \neq 0$.

Demonstration for the binomial. Write $X = B_1 + \cdots + B_n$ with independent indicators of success. B_i checks $E(B_i) = p$, $E(B_i^2) = p$ and $V(B_i) = p - p^2$. By linearity and additivity under independence, $E(X) = np$, $V(X) = np(1-p)$.

Sample and mean

A size sample n of a law is a family of n variables of the same law.
have expectation μ and variance σ^2, we note $S_n = \sum_{i=1}^{n} X_i$ and $M_n = \frac{S_n}{n}$.

Sum and average

$$E(S_n) = n\mu, \quad V(S_n) = n\sigma^2, \qquad E(M_n) = \mu, \quad V(M_n) = \frac{\sigma^2}{n}, \quad \sigma(M_n) = \frac{\sigma}{\sqrt{n}}.$$

To divide the standard deviation of the average by 2, multiply the size by 4. The variance of the sum increase; the average is decreasing.

Concentration

Inequality of Bienaymé-Chebychev

For a variable of expectation μ, variance σ^2, and $\delta > 0$,

$$P(|X-\mu| \geq \delta) \leq \frac{\sigma^2}{\delta^2}.$$

A limit of more than 1 remains true, but does not provide additional information.
may be replaced by 1.

Proof in the finished frame. In $V(X) = \sum_x (x-\mu)^2 P(X=x)$, only keep the values checking $|x-\mu| \geq \delta$. Each square is then at least δ^2. So $V(X) \geq \delta^2 P(|X-\mu| \geq \delta)$.

Law of large numbers

Applied to the average of a sample,

$$P(|M_n - \mu| \geq \delta) \leq \frac{\sigma^2}{n\delta^2} \xrightarrow[n\to\infty]{} 0.$$

This is the law of the large numbers in the given framework: for a specified accuracy, the probability of at least equal to this precision tends to zero. This result does not say that each achievement is close to μ in a monotonous manner.

Select a sufficient size

To guarantee $P(|M_n - \mu| < \delta) \geq 1 - \alpha$, just choose

$$n \geq \frac{\sigma^2}{\alpha\delta^2}.$$

Take the ceiling. It is the smallest satisfying integer this bound , not necessarily of minimum size for the exact probability.

For a frequency of success, $\mu = p$ and $\sigma^2 = p(1-p) \leq \frac{1}{4}$. If p is unknown, $n \geq \frac{1}{4\alpha\delta^2}$ is a guarantee uniform.

From individual law to average

14.1 Describe a non-binomial variable

First use: S62.

Example resolved · A repair cost

A cost X in euros takes values 0, 20 and 50 with probabilities 0 ,5, 0,3 and 0,2. $E(X) = 16$, $E(X^2) = 620$, therefore $V(X) = 620 - 16^2 = 364$ in square euros. $\sqrt{364} \simeq 19{,}08$ Euros. Average 16 is none of the possible values of X; it describes an average theoretical on many achievements.

A1 · Immediate application X takes values 1, 2, 4 with probabilities 1 /2, 1/4, 1/4. Check the Law and calculate expectation, variance and standard deviation.

14.2 Remission also changes dispersion

First use: S62.

Property · Affine processing

For $Y = aX + b$, $E(Y) = aE(X) + b$ and $V(Y) = a^2V(X)$. The standard deviation is $|a|\sigma(X)$. Adding a constant translates all values without changing their dispersion.

Example resolved · Cost after remission

For the previous cost, a discount of 20% and then 5 euros of costs gives $Y = 0{,}8X + 5$. Thus $E(Y) = 17.8$ and $V(Y) = 0{,}64 \times 364 = 232.96$. The variance is neither $0.8V(X) + 5$ or $0.8\,V(X)$.

A2 · Partial guidance For the law of the previous application, calculate E and V of $Y = 3X - 2$ and then $Z = -X$. Explain why multiply by -1 does not change the variance.

14.3 Independence intervenes in the variance

First use: S64.

Property · Sum and average

For variables $X_1, \ldots, X_n$ of the same expectation μ and even variance σ^2, average M_n a for expectation μ. If independent, $V(M_n) = \sigma^2/n$ and $\sigma(M_n) = \sigma/\sqrt{n}$. Linearity expectation does not require independence.

Demonstration. With $S_n = \sum X_i$, linearity gives $E(S_n) = n\mu$. Under independence, additivity Allowed variances give $V(S_n) = n\sigma^2$. Multiply by 1 /n for the average multiply expectation by 1 /n and variance by 1 /n^2. For Bernoulli, E=p and $V = p(1-p)$; the binomial, sum of n Bernoulli independent, therefore has E=np and $V = np(1-p)$. □

Example resolved · Average of 100 costs

Per cent independent costs of the same law, of expectation 16 and of variance 364, the total has for expectation 1600 and variance 36400. The average cost has for expectation 16 and variance 3.64, hence a Standard deviation of around EUR 1.91.

A3 · Counterexample X is 0 or 1 with probability 1 /2 each. Compare variances of $(X_1 + \cdots + X_n)/n$ for independent copies, then for $X_1 = \cdots = X_n = X$.

14.4 Sufficient guarantee, not an exact probability

First use: S73.

Property · Bienaymé-Chebychev

For $\delta > 0$, $P(|X - \mu| \geq \delta) \leq V(X)/\delta^2$. For an independent average, $P(|M_n - \mu| \geq \delta) \leq \sigma^2/(n\delta^2)$. Any bound may be reduced to 1 if it exceeds it.

Example resolved · Size guaranteed by the bound

For a probability success frequency 1 /2, $V(M_n) = 1/(4n)$. We want a strict deviation less than 0 ,1 with a probability at least 0 ,95. It is sufficient to have $25/n \leq 0{,}05$, so $n \geq 500$. This gives the smallest integer satisfying this sufficient inequality; it does not prove that 499 observations actually fail.

A4 · Immediate application A variable has a variance of 9. independent guarantees, by this bound, an average deviation strictly less than 0 ,5 with one Like at least 0.99?

A5 · Compare two informations For $X \sim \mathcal{B}(10; 1/2)$ and $M = X/10$, calculate exactly $P(|M - 1/2| \geq 0{,}3)$ and compare to the Tchebychev bound. Work with the exact entire bounds.

14.5 Several samples, several averages

First use: S74 and S77.

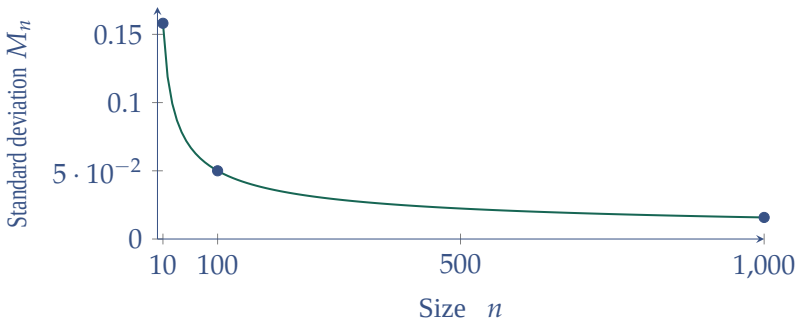


Figure 14.1. Theoretical curve 1/ $(2\sqrt{n})$: multiply n by 4 divides standard deviation by 2, not by 4.

Example resolved · What is compared

With n=100, the theoretical standard deviation of a probability-success frequency 1/2 is 0.05. series of 1000 experiments of 100 tests produces 1000 averages. a series of averages compared to 0.05; the 100 000 individual results have another dispersion.

A6 · TP and interpretation With the program provided, simulate 1,000 samples independent of sizes 10,100 and 1000 for a Bernoulli parameter 1 /2. For each size, compare the standard deviation averages to 1 $/(2\ \overline{n})$ and count deviations of at least 0 ,1. Why a single trajectory Does it not answer this question?

A7 · Refuter "The law of large numbers says the frequency is approaching expectation give a short sequence of results that temporarily removes the frequency of 1/2, then reformulate the result correctly.

Wait a minute at the courthouse. Make a sentence containing "accuracy", "risk" and "sufficient size", then distinguish this sentence from certainty for a particular sample.

Delayed review · S90. In a topic of synthesis, identify first the law or variance of the average, before choosing the threshold.

Research activity

Fluctuations of frequency · 40 min.

We independently launch a balanced piece. $n \geq 1$, note F_n the frequency of « pile » among the n first shots.

1. Write a program that simulates 1000 throws, then represent points $(n; F_n)$ for $1 \leq n \leq 1000$. The gap to $\frac{1}{2}$ Does it necessarily decrease at each pitch?
2. Express $E(F_n)$ and $V(F_n)$ according to the n.
3. With the help of Bienaymé-Chebychev, increase $P(|F_{1000} - \frac{1}{2}| \geq 0{,}05)$.
4. Determine an Integer n sufficient to ensure $P(|F_n - \frac{1}{2}| < 0{,}05) \geq 0{,}99$.
5. Explain what the simulated trajectory shows and what the inequality achieved guarantees. Is it proof of this guarantee?

Progressive exercises

E1 · Direct application. Let X a random variable such as $E(X) = 3$ and $V(X) = 2$. Installation $Y = 4X - 1$. Calculate the expectation, variance and standard deviation of Y.

E2 · Application and justification. Random variables X and Y are independent and verify $E(X) = 2$, $E(Y) = 5$, $V(X) = 1$ and $V(Y) = 4$. Calculate $E(2X - Y)$ and $V(2X - Y)$.

E3 · Method to be organised. We consider 64 random variables independent of the same law, expectation Calculate the expectation and standard deviation of their average.

E4 · Choice of strategy. Let X a random variable such as $E(X) = 100$ and $V(X) = 9$. With the help of Inequality of Bienaymé-Chebychev, give an increase of $P(|X - 100| \geq 6)$.

E5 · Synthesis or demonstration. Note M_n the average of n random variables independent of same law, expectation μ 4. Determine a size n sufficient to ensure $P(|M_n - \mu| < 0{,}2) \geq 0{,}95$.

Work in the manual or annal

Baccalaureate, Métropole, 17 June 2025: Exercise 1, question 6 (p. 3). At home after S77. Read the the context of question 5 and use the values of expectation and variance accepted in question 6. The subject is available here: official annual . Use the data and instructions of this yearnal.

Duty at home

DM14 · Quality of an estimate · 55 min.

Repeat $n \geq 1$ Bernoulli tests independent of the same parameter $p \in [0;1]$, unknown. note F_n the frequency of success.

1. Demonstrate that $E(F_n) = p$ and $V(F_n) = \frac{p(1-p)}{n}$.

2. Show that $p(1-p) \leq \frac{1}{4}$ for every $p \in [0;1]$.

3. Deduce a size n sufficient to ensure $P(|F_n - p| < 0{,}02) \geq 0{,}95$, whatever the value of p.

4. We now know that $p \leq 0{,}1$. Determine a more precise increase of $p(1-p)$then improve the sufficient size obtained in question 3.

5. On this question alone, the n results are all equal to the same Bernoulli variable B parameter p. Calculate F_n and its variance. Explain why the previous warranty does not apply More.

QCM A · transformations, amounts and averages

After S64; QCM B after S77.

For each question, only one proposal is accurate.
consult the correction.

1. $E(X) = 3$ and $Y = 2X - 1$. Then $E(Y)$ is valid for:

A. 5 B. 6

C. 2 D. 8

2. $V(X) = 4$ and $Y = -3X + 5$. Then $V(Y)$ is valid for:

A. -12 B. 17

C. 36 D. 41

3. For n independent variables of variance σ^2, the variance of their average is:

A. $n\sigma^2$ B. σ^2/n

C. σ^2/n^2 D. σ^2

4. Equality $E(X+Y) = E(X) + E(Y)$:

A. demands independence B. requires $X = Y$

C. is false D. applies without independence for variables finished

5. X is 0 or 2 with probabilities 1/2 each. Its variance is:

A. 0 B. 2

C. 1 D. 4

QCM B · scope of warranty

For each question, only one proposal is accurate.
consult the correction.

6. For $a > 0$, Bienaymé-Chebychev gives:

A. $P(|X - E(X)| \geq a) \leq V(X)/a^2$ B. $P(|X - E(X)| \geq a) = V(X)/a^2$

C. $P(|X - E(X)| \geq a) \geq V(X)/a^2$ D. $P(X \geq a) \leq E(X)/a^2$

7. For a Bernoulli frequency, $p = 1/2$, the bound for $P(|F_n - p| \geq 0{,}1)$ is 25 $/n$. To give it back not more than 0.05, it is sufficient to:

A. $n \geq 25$ B. $n \geq 100$

C. $n \geq 250$ D. $n \geq 500$

8. A Chebychev bound equal to 2 can only be deduced from:

A. a probability equal to 2 B. Trivial bound 1

C. impossibility D. a zero probability

9. The law of large numbers states that, for all $a > 0$ fixed:

A. the frequency becomes exactly equal to p B. Each new test reduces the error

C. $P(|F_n - p| \geq a) \to 0$ D. the observations are constant

10. If $X_1 = \cdots = X_n = X$ and $V(X) > 0$, the variance of their average is:

A. $V(X)$ B. $V(X)/n$

C. 0 D. $nV(X)$

True or false · expected justification. For each statement, indicate whether it is true or false, and then write evidence or a counter-example.

1. A sufficient size given by Chebychev is necessarily the actual minimum size.

2. Standard deviation of $aX + b$ is $|a|\sigma(X)$.

3. Independence may be omitted in the calculation of variance of an average.

Corrected MCQs

1. **A.** By linearity, $E(Y) = 2E(X) - 1 = 5$. The constant is subtracted once.
2. **C.** The variance is multiplied by $(-3)^2 = 9$, and a translation does not change it. negative.
3. **B.** The sum has variance $n\sigma^2$, then division by n multiplies variance by 1/ n^2.
4. **D.** The linearity of expectation does not require independence. additional conditions.
5. **C.** $E(X) = 1$, $E(X^2) = 2$, so $V(X) = 2 - 1^2 = 1$.
6. **A.** It's an increase in a gap to expectation, not an equality or a point bearing directly on X.
7. **D.** Solve 25/ $n \leq 0.05$ with $n > 0$ given $n \geq 500$. It is a sufficient size according to this bound.
8. **B.** A probability is always at most 1; the increase 2 does not give any improvement.
9. **C.** This is a convergence in probability, not a decrease in error at each observation.
10. **A.** The average is exactly X. Repeating the same variable does not create new observations independent.

True/false. False: the increase can be large. True: take the root of $a^2V(X)$. False: question 10, where variance does not decrease.

Corrected applications and support

Application 1. Sum of probabilities 1. E=2; $E(X^2) = 11/2$, therefore $V = 3/2$ and $\sigma = \sqrt{6}/2$.

Application 2. $E(Y) = 4$, $V(Y) = 27/2$; $E(Z) = -2$, $V(Z) = 3/2$. The square of the deviations to the average does not change under sign inversion.

Implementation 3. Independence: variance 1 $/(4n)$. Identical copies: the average is X, so variance 1/4 whatever n. Repeating the same measurement does not create an independent sample.

Implementation 4. The bound is 9/ $(0{,}25n) = 36/n$. Require $36/n \leq 0.01$ given $n \geq 3600$.

Application 5. The event is $X \leq 2$ or $X \geq 8$. By symmetry, its probability is 2 $(1 + 10 + 45)/2^{10} = 7/64 \simeq 0{,}109375$. The bound is $(1/40)/(9/100) = 5/18 \simeq 0{,}2778$, so it's correct But less precise.

Implementation 6. Theoretical standard deviations are about 0 ,1581, 0,05 and 0,01581. Simulated outputs fluctuation: a set of means is compared, not the successive values of a single average. Empirical proportions are not demonstrations. The reproducible results file gives the values obtained with the seed retained.

Implementation 7. Results 0.1 give frequency 1 /2 ; add 1 gives 2 /3, further from 1 /2. For the probability of a deviation at least equal to that accuracy tends to 0 when n This does not impose a monotonous rapprochement on each achievement.

Reading the guarantees

Surround the strict sign of the central event and the broad sign of its complement. Compare a computable probability and a universal limit. Do not present the simulation as evidence of the law of large numbers.

Conduct and correction

Benchmarks for the teacher

The DS5 and the second white precedes S73: do not ask for Bienaymé-Chebychev nor law
for variance, make surround the hypothesis of independence.
concentration, distinguishing exact probability and increase, then sufficient rank and minimum rank
A simulation is not evidence and an observed average does not approach
Of course, every time you try again.

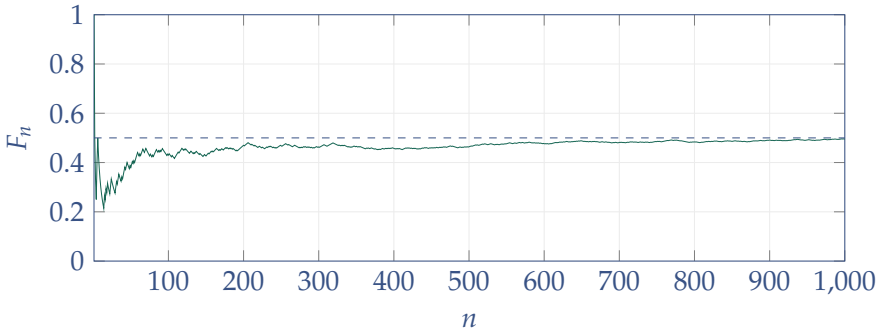
Corrected activity

1.

```
From random import Random

def trajectories_piece(N=1000, seed=2026):
    rng = Random(grain)
    batteries, frequency = 0, []
    for n in rows (1, N+1):
        batteries += rng.random() < 0.5
        frequency.append(cells/n)
    return frequency
```

Trace points (n, F_n) with the list obtained. After two tests giving a battery and one face, $F_2 = \frac{1}{2}$. Whatever the third result, F_3 value $\frac{1}{3}$ or $\frac{2}{3}$: difference at $\frac{1}{2}$ Stabilisation increases. is not a monotonous decrease of the deviation on each trajectory.



A simulated trajectory, seed 2026; the right is $y = \frac{1}{2}$.

2. Note: B_i I'm the battery-indicator. $F_n = \frac{B_1+\cdots+B_n}{n}$. Each B_i has expectation $\frac{1}{2}$ and variance $\frac{1}{4}$. By linearity, $E(F_n) = \frac{1}{2}$; by independence and change of scale, $V(F_n) = \frac{n(\frac{1}{4})}{n^2} = \frac{1}{4n}$.

3. Bienaymé-Chebychev gives $P(|F_{1000} - \frac{1}{2}| \geq 0{,}05) \leq \frac{1}{4\times 1000\times 0{,}05^2} = 0{,}1$. It is a bound higher, not the exact probability.

4. To ensure a complementary probability of up to 0 ,01, it is enough to $\frac{1}{4n\times 0{,}05^2} \leq 0{,}01$. This is equivalent to to $n \geq 10000$. $P(|F_n - \frac{1}{2}| < 0{,}05) \geq 0{,}99$.

5. The simulation illustrates a finished realization; other seeds give other trajectories. establishes a guarantee of probability for any sample meeting the assumptions of independence and common law.

E1. The affine transformation gives $E(4X-1) = 4E(X) - 1 = 11$ and $V(4X-1) = 4^2 V(X) = 32$. The standard deviation is $\sqrt{32} = 4\sqrt{2}$; the lag -1 does not change dispersion.

E2. By linearity, $E(2X-Y) = 2E(X) - E(Y) = 4 - 5 = -1$. The independence of X, Y allows add the variances of 2 X and $-Y$: $V(2X-Y) = 4V(X) + V(Y) = 4 + 4 = 8$.

E3. The average of 64 independent variables of the same law retains expectation 10. value $\frac{4^2}{64} = \frac{1}{4}$, so its standard deviation is $\frac{1}{2}$.

E4. Bienaymé-Chebychev with $\delta = 6$ given $P(|X-100| \geq 6) \leq \frac{V(X)}{6^2} = \frac{9}{36} = \frac{1}{4}$. This bound does not does not give an exact equality or value.

E5. We have $P(|M_n - \mu| \geq 0{,}2) \leq \frac{4}{n \times 0{,}2^2} = \frac{100}{n}$. For this bound to be not more than 0 ,05, it is enough to $n \geq 2000$. The complement then has a probability of at least 0.95.

Corrected from DM14

1. Note: B_i independent indicators of success. $F_n = (\frac{1}{n}) \sum_{i=1}^{n} B_i$, $E(B_i) = p$, $V(B_i) = p(1-p)$. So

$$E(F_n) = \frac{1}{n}\sum_{i=1}^{n} p = p, \qquad V(F_n) = \frac{1}{n^2}\sum_{i=1}^{n} p(1-p) = \frac{p(1-p)}{n}.$$

2. $p(1-p) = p - p^2 = \frac{1}{4} - (p - \frac{1}{2})^2 \leq \frac{1}{4}$. Equality is achieved for $p = \frac{1}{2}$.

3. For every $p \in [0,1]$, $P(|F_n - p| \geq 0{,}02) \leq \frac{1}{4n \times 0{,}02^2}$. It is sufficient that this bound is not more than 0,05, i.e. $n \geq 12500$. By addition, this size guarantees the required probability, uniformly in p.

4. On $[0,0{,}1]$, function $p \mapsto p(1-p)$ is increasing as its derivative 1 $- 2p$ is positive. $p(1-p) \leq 0{,}1 \times 0{,}9 = 0{,}09$. It is now sufficient to $n \geq \frac{0{,}09}{0{,}05 \times 0{,}02^2} = 4500$. These sizes are sufficient within the meaning of the mark-up, not necessarily minimum for the exact probability.

5. If all tests copy the same variable B, then $F_n = \frac{B + \cdots + B}{n} = B$. So $V(F_n) = p(1-p)$, without division by n. The addition of variances was permitted only through independence; The hypothesis disappears here. $< p < 1$, increase the number of copies does not bring any new information.

Corrected work in the manual or an anal

M_N is the average number of universal donors per sample of 100 people. Linearity gives $E(M_N) = 7.14$; independence gives $V(M_N) = \frac{6{,}63}{N}$.

$$P(|M_N - 7{,}14| \geq 0{,}14) \leq \frac{6{,}63}{0{,}14^2 N}.$$

To make this bound equal to or less than 0.05, just take

$$N \geq \left\lceil \frac{6{,}63}{0{,}05 \times 0{,}14^2} \right\rceil = 6766.$$

For additional information, $P(7 < M_N < 7{,}28) \geq 0{,}95$.